



Teaser This review examines the efficiency of therapeutics enhancing the clearance of bacteria persisting inside host cells. The proposed novel therapeutics target host autophagy in conjugation with existing treatments to enhance the removal of persistent intracellular bacterial pathogens.

Therapeutic strategies against autophagic escape by pathogenic bacteria

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Growing multidrug-resistant (MDR) strains of various infectious bacterial species are hindering research aiming to eliminate such infections.

Q8 During a bacterial infection, the host response eliminates the pathogen via fusion of the endocytic vesicles with lysosomes, called xenophagy. However, MDR bacteria have evolved strategies to escape xenophagy. In this review, we propose novel therapeutics for overcoming such escape, including chimeric antibiotics, nanoformulations for the induction of autophagy in infected cells, and small interfering (si)RNA-mediated silencing of genes to inhibit the host–pathogen interaction. We also discuss the role of combinations of antibiotics showing synergy, the administrative routes of differentially capped nanoparticles (NPs), and the use of different types of nanoformulations for eliminating pathogenic bacteria from the host.

Introduction

The widespread increase in the origin of MDR bacteria strains can be attributed to the erroneous use of antibiotics. This has led to increases in various bacterial infections that are unresponsive to currently available treatments [1]. Searching for new therapeutics against such pathogens has become challenging, especially given a significant decline in the research and development of antibiotics and pharmaceuticals [1]. Subsequently, targeting the natural host responses against these pathogens has become a promising strategy for the development of novel therapeutics. Of the various host responses to pathogens, autophagy is one of the most common, responsible for processing and removing impaired organelles and misfolded proteins [2]. Autophagy induced by microbes involving the trafficking of intracellular bacteria in the lysosomes is known as xenophagy [3]. One reason why this process is considered to have potential as a therapeutic target is that it is a much-conserved process across organisms, from single-celled organisms to mammals. Xenophagy involves the isolation and formation of double-layered membranes that swallow and transport their cargo to lysosomes, where they are then destroyed [4]. extensive studies on host–pathogen interactions have highlighted the benefits of autophagy induction as a strategy to treat bacterial infection [5,6]. However, recent studies also revealed manipulative pathways used by multiple bacterial species to avoid encountering host autophagy [7]. In this review, we provide an overview of the primary mechanism of autophagy in the host to highlight probable targets for new therapeutics. We also discuss the use of host autophagic responses for

developing novel therapeutics to help maintain health and diseases [8]. A variety of options have been suggested to overcome MDR pathogens, including conjugation of plant secondary metabolites with existing antibiotics, a combination of the functional domains of last-resort antibiotics for synergistic effects against bacteria, and the use of nanoformulations for effective drug delivery. The focus of each lies in developing those therapeutics that could effectively eliminate bacterial infections without the risk of developing new, more severe types of bacterial resistance.

Battle for survival: bacterial pathogen-induced host autophagy

Understanding the basic mechanism of autophagy is essential before proceeding to possible therapeutics. The constitutive expression of autophagy is essential for generalized homeostasis, because it regulates the levels of cellular organelles and proteins [9]. In addition, whenever the cell encounters any stress, infection, or drug treatment, there is a change in the basal level expression of autophagy-related genes [9]. Thus, research is focusing on uncovering the fundamental molecular mechanisms involved in autophagy.

Autophagy induction begins with the recognition of nutrient-starved, damaged, and lysed cell organelles [10]. These are sensed by Unc-51-like kinase 1 (Atg1 in yeast), a potent downregulator of the target of rapamycin complex 1 (TORC1) with a role in the growth and maintenance of cells in nutrient-rich conditions [11]. The ULK1 complex comprises ULK1, ATG13, FAK family kinase-interacting protein of 200 kDa (FIP200), and ATG101, and is vital for the initiation of phagophore membranes [12], given that, during starvation conditions, TORC1 is unable to phosphorylate the ULK1 complex, and it is released into the system. The nucleation of the vesicle is then mediated by the recruitment of the class III phosphatidylinositol 3-kinase complex, including beclin 1, PI3K(Vps34 in yeast), VPS15 (PIK3R4), and ATG14L [13]. The elongation of the autophagosome is controlled by ubiquitin-like conjugation factors, the ATG12–ATG5 ubiquitin-like-conjugate, activated by the E1 and E2-like enzymes ATG7 and ATG10 [14]. By contrast, ATG12–ATG5 complexes with ATG16L1 are incorporated into the expanding phagophore [15]. They also simultaneously recruit another ubiquitin-like conjugate, LC3 (ATG8 in yeast), with phosphatidylethanolamine (PE) via the action of ATG4 at the C-terminal-exposed glycine of LC3 [16]. LC3 is then recruited

to the cargo because the cargo/pathogen-containing vesicles have an LC3-interacting domain (LIR) [17]. Ubiquitinated cargo is then recognized via ubiquitin-binding protein adapters (p62, NBR1, NDP52, and optineurin), which have both a ubiquitin-binding site and LIR domain [11]. The resulting mature phagosome is then ready to fuse with the lysosome to initiate the degradative process. For this purpose, various proteins are essential, including Rab7 (GTPase) [18], SNARE protein syntaxin 17 on autophagosome, VAMP8 on the lysosomal membrane, and some LAMP proteins [19]. Once an appropriate interaction has been established between the endosome and lysosome, the lysosomal hydrolases degrade the inner membrane of the autophagosome, followed by its cargo [11].

There are various receptors specific for each type of pathogen that enable their internalization inside host cells. For example, *Salmonella typhimurium*, after being internalized, occurs in *Salmonella*-containing vacuoles. The type III secretion system creates pores in these vacuoles and, as a result, the bacteria are released into the cytosol [20]. These bacteria are ubiquitinated and, therefore, the binding of protein adapters (p62, NDP52, and NBR1) facilitates the formation of LC3-coated phagosomes [21]. Similarly, in *Shigella flexneri* and *Mycobacterium tuberculosis*, recognition occurs via the same adapter molecules [22]. For *Listeria monocytogenes*, the ubiquitination process is regulated by Smurf1 and inhibits any further proliferation in host cells [23]. During *L. monocytogenes* infection, the bacterium secretes a pore-forming effector, LLO, to create starvation conditions in the host and, thus, inhibit mTOR [24]. *Streptococcus pyogenes* also creates pores in the endosomal membrane via the release of Streptolysin O, and the bacteria are then recognized by the adapter proteins (p62 and NDP52) [25]. In *Acinetobacter baumannii*, the mode of autophagy induction is attributed to the upregulation of the levels of transcription factor EB in host cells [26]. TF-EB is a gene involved in the biogenesis of lysosomal genes; thus, the increasing number of lysosomes is one of the main reasons for bacterial clearance from the host system. Various proteins are involved in the induction of autophagy in *Acinetobacter* infections; for example, Omp33-36 is a porin that induces apoptosis and, hence, also modulates autophagy [27]. Another crucial factor is Omp-A, which causes incomplete autophagy via the MAPK/JNK signaling pathway [28]. It also regulates the expression of c-jun and c-fos involved in promoting the apoptotic pathway. Specific host–pathogen interactions are detailed in Table 1.

TABLE 1

Effector molecules of pathogens and their specific host interacting proteins

Bacteria	Effector molecule (bacterial secretions)	Host interacting protein	Refs
<i>Acinetobacter baumannii</i>	TF EB, Omp33-36	Lysosomal biogenesis, MAPK/JNK pathway	[27,28]
<i>Legionella pneumophila</i>	T4SS protein, RavZ, and Lpg1137	Disassociation of LC3 and PE; degradation of Stx17	[32]
<i>Listeria monocytogenes</i>	LLO	Inhibition of mTOR	[31]
<i>Mycobacterium marinum</i>	ESx-1	Does not enter LC3 compartments	[34]
<i>Mycobacterium tuberculosis</i>	Esx-1 secretion system and SapM	P62, NDP52, NBR1 UBQLN1; inhibit Rab5-Rab7 exchange	[30]
<i>Neisseria gonorrhoeae</i>	Type IV pilus	CD46-cyt1 pathway	[37]
<i>Salmonella typhimurium</i>	Type III secretion system	Ubiquitin protein adapters (p62, NDP52, NBR1, and Optineurin LRSAM1)	[29]
<i>Shigella flexneri</i>	LPS	MAPK pathway	[36]
<i>Staphylococcus aureus</i>	α -hemolysin	LAMP-1	[33]
<i>Streptococcus pyogenes</i>	Streptolysin O	Recognition by p62, NDP52	[25]
<i>Yersinia pseudotuberculosis</i>	T3SS	P62, NDP52	[35]

The evolution of autophagic escape strategies in pathogenic bacteria

The autophagic machinery is vital for bacterial infection elimination. However, these pathogenic bacteria manipulate the autophagic process to enhance their chances of survival. It is possible to manipulate the host machinery at different stages, such as the initiation step of autophagy, maturation of autophagosomes, recruitment of autophagic adapter proteins, and the process of autophagosome and lysosome fusion (Fig. 1). Various molecular-level studies have reported the primary strategies of pathogens to escape xenophagy. For example, *S. flexneri*, the causative agent of Shigellosis, has developed resistance against the fluoroquinolones and co-trimoxazole [29]. It also escapes autophagy via targeting the Rab1 GTPase via secretion of effector/virulence factors, such as VirA, inhibiting host autophagy initiation. Similarly, strains of *S. typhimurium* that show resistance to antibiotics, such as streptomycin and chloramphenicol [30], inhibit the initiation of autophagy via the SseF effector (Table 2). MDR *M. tuberculosis* resistant against rifampicin and isoniazid [31] inhibit autophagy induction by suppressing the JNK-ROS pathway of the host via its effector molecule Eis. *L. monocytogenes* secrete phospholipases PlcA and PlcB, which inhibit the recruitment of PI3P [24]. *Legionella pneumophila* causes community-acquired pneumonia and has evolved resistance against ceftriaxone, rifampicin, and azithromycin [32]. It survives autophagy via deconjugation of LC3-PE by the secretion of RavZ from T4SS; hence no autophagosome formation occurs [33]. *S. pyogenes*, a causative agent of toxic shock syndrome and the leading cause of thousands of infections, has evolved antibiotic resistance even against the last resort antibiotic (i.e., β -lactams) [34]. It secretes an effector molecule, SpeB, which degrades the adapter proteins and hinders the recognition of the bacteria after ubiquitination [35]. By contrast, *Staphylococcus aureus* inhibits the maturation of autophagosomes through its effector molecules Hla and IsaB, and creates a replicative niche in the autophagosome itself [36]. The induction of autophagy kills *Neisseria gonorrhoeae* via the CD46-cyt1 pathway, but because of some unknown factor, this pathogen manipulates autophagy initiation [37]. Hence, these various steps could prove useful targets for developing novel therapeutics for treating bacterial infections. Table 2 highlights the use of various toxins/effector molecules used by bacterial pathogens to escape host autophagy.

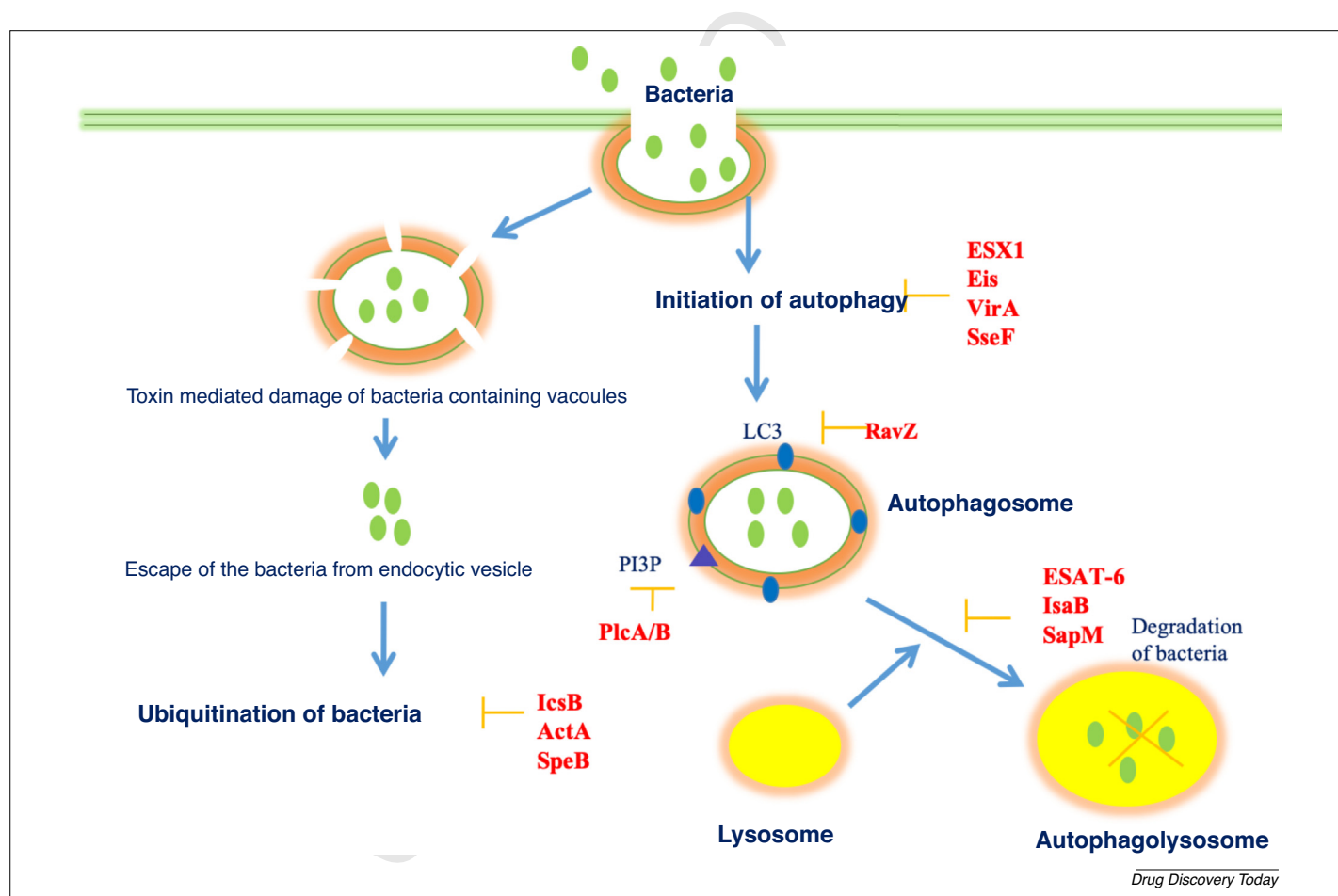


FIGURE 1

Q1 Hijacking of the host autophagy by bacteria at different stages of infection. Initiation of autophagy is blocked by ESX-1 (ESAT-6-like protein of *Mycobacterium marinum*), Eis (Enhanced intracellular survival protein of *Mycobacterium tuberculosis*), VirA (Cysteine protease-like VirA of *Shigella flexneri*), SseF (Secretion system effector of *Salmonella typhimurium*). The ubiquitination of bacteria is blocked by IcsB (Virulence protein IcsB of *S. flexneri*), ActA (Actin assembly-inducing protein of *Listeria monocytogenes*), and SpeB (streptococcal pyrogenic exotoxin of Group A *Streptococcus*). ESAT-6 (6-kDa early secretory antigenic target of *M. tuberculosis*), IsaB (Immunodominant surface antigen B of *Staphylococcus aureus*), and SapM (Secreted acid phosphatase of *M. tuberculosis*) have been found to inhibit the fusion of autophagosomes with lysosomes.

TABLE 2

Escape mechanisms used by bacteria to manipulate host autophagy

Bacteria	Escape strategy (virulence factors secreted by bacteria)	Host target to manipulate autophagy	Refs
<i>Enterococcus faecium</i>	PerA, glycosaminoglycans	Unknown	[44]
Group A <i>Streptococcus</i>	Streptolysin O, SpeB	p62, NDP52	[51]
<i>Legionella pneumophila</i>	RavZ	LC3-II	[32]
<i>Listeria monocytogenes</i>	ActA	Actin	[45]
<i>Listeria monocytogenes</i>	Phospholipases PlcA and PlcB	PI3P	[49]
<i>Mycobacterium marinum</i>	ESX-1	TORC1	[52]
<i>Mycobacterium tuberculosis</i>	Eis	JNK-ROS	[47]
<i>Neisseria gonorrhoeae</i>	Unknown	Degradation of CD46-cyt1 and LAMP	[50]
<i>Salmonella typhimurium</i>	SseF, SseG	Rab1A GTPase	[46]
<i>Shigella flexneri</i>	VirA IcsA/B	Rab1 Atg5	[22]
<i>Staphylococcus aureus</i>	Hla, IsaB	Inhibit autophagosome maturation	[48]
<i>Streptococcus pyogenes</i>	SpeB	Degradation of p62, NDP52, and Optineurin	[43]

Therapeutic strategies against the escape of pathogens

From our understanding of the fundamental processes involved in autophagy and the various escape pathways used by different bacterial species, it is possible to devise mechanisms to target and stop the infections in their early stages. Here, we discuss the use of nanoformulations in conjugation with plant metabolites, the use of siRNA to silence receptors of bacteria on host cells, and the designing of chimeric molecules based on previous understanding of the mechanism of action of currently available antibiotics. Various pathogens, such as *A. baumannii*, *Shigella*, and *Salmonella*, which hijack the autophagy of the host, are the main targets for the development of therapeutics. Given that different pathogens acquire different mechanisms to overcome host immunity, the targeted therapeutics should be specific to them. For example, smart chimeric peptides targeting lipopolysaccharide (LPS) and fusion of the functional domains of the currently available antibiotics have been used to eliminate *Escherichia coli* and *A. baumannii*, respectively. Similarly, the use of silver NPs (AgNPs) to target *S. aureus* and their efficacy after fusion with common antibiotics and various other metal NPs is also a promising approach to target autophagy escapers. The most important aim is to increase the autophagic flux of the host for more enhanced clearance of the pathogens. This could be done effectively by conjugating autophagy inducers with the antibiotics, as discussed later. Figure 2 provides overview of the interactions that could be targeted to eliminate bacterial infections.

Chimeric antibiotics against MDR bacteria

Continuous administration of antibiotics at higher concentrations can also cause adverse effects during pregnancy, initiation of disease, and the emergence of new potent strains of pathogens with no available treatment [38]. Here, we focus on the use of therapeutics that come with minimum risks of adverse effects. For example, the functional domains of murepavadin and polymyxin B1 were combined to form chimeric antibiotics and analyzed for their antibacterial and cytotoxic effects against Gram-negative bacteria [39]. The LPS and outer membrane protein were the targets of this chimeric molecule, increasing the effective target sites for the elimination of the bacteria. The effectiveness of these antibiotics was also tested against *A. baumannii*, a Gram-negative bacteria of the ESKAPE (*Enterococcus faecium*, *S. aureus*, *Klebsiella*

pneumoniae, *A. baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* spp.) group [39]. Subtractive proteomics helps to identify the virulence factors and genes essential for the survival of the pathogen [40]. Hence, identifying these will reveal genes that could be targeted with chimeric antibiotics. Alternatively, *Shigella* and *Salmonella* can be targeted using substances that include the functional domains of β -lactam antibiotics [41].

A study involving MDR *E. coli* revealed LPS as an authentic drug target because it neutralizes drugs used for treatment and causes septic shock [42]. Researchers designed smart chimeric peptides (SCPs), A6 and G6, including combinations of LPS-targeting peptide, LBP14, and killing domain, N6 [43]. Linkers were used to combine these two domains to enhance the effectiveness of the peptides because they have a role in the intrinsic biological properties (folding and flexibility) of the peptide [44]. Similarly, endotoxins such as T3SS, ESAT-6, Eis, outer membrane proteins, and other factors involved in bacterial escape from autophagy could be targets for the development of therapeutics. The benefit of using chimeras is that it minimizes the risk of bacteria developing resistance, and the MICs are significantly less compared with currently existing antibiotics [43].

Nanoformulation and autophagy inducers against persistent bacteria

NPs are well known for their antibacterial activity, and coating them with the plant extracts enhances their stability and antibacterial activity [45]. Plants are traditionally known for their antibacterial activities and it is relatively inexpensive to screen the conjugation of their secondary metabolites with NPs. For example, the coating of NPs with chitosan imparts a positive charge to them, enabling their interactions with, and disruption of, negatively charged microbial membranes. It has also been reported to inhibit mRNA and protein synthesis [46].

Polyvinylpyrrolidone (PVP)-capped AgNPs effectively kill carbapenem-resistant bacterial strains and decrease the viability of intracellular bacteria by up to 80% [45]. Some studies also confirmed a significant decrease in surface-interacting and engulfed/internalized bacteria after treating infected A-549 cell lines with PVP-capped AgNPs [45]. The NPs are internalized via Scavenger receptor-mediated endocytosis [47] and, hence, this common endocytic pathway could be utilized to target persistent bacterial cells. AgNPs have also shown significant antibacterial activity

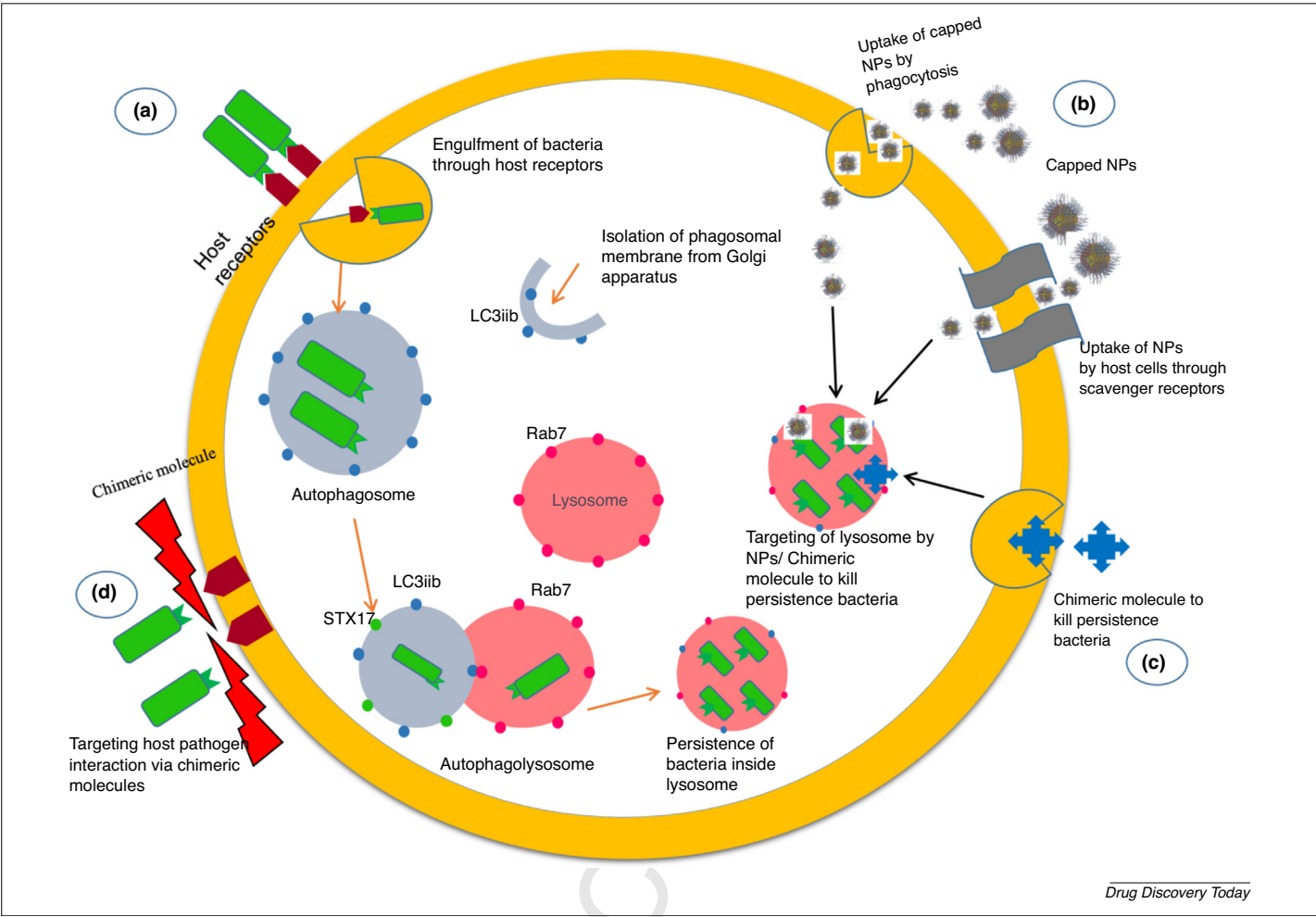


FIGURE 2 Possible therapeutic options against persistent bacterial species. **(a)** general autophagic process; **(b)** treatment of bacterial infections via differentially capped nanoparticles (NPs); **(c)** use of chimeric inhibitors to kill persistent bacteria; and **(d)** use of chimeric molecules that inhibit host–pathogen interactions.

against the ESKAPE group and drug-resistant strains of bacteria, including *S. aureus* [48]. Studies have also revealed the increased synergistic effects when the NPs were used in conjugation with antibiotics such as penicillin G, amoxicillin, erythromycin, clindamycin, and vancomycin [49].

Table 3 details NPs showing improved synergy when administered in combination with antibiotics. The combination of gold (Au)-Ag alloy NPs with kanamycin showed significant results even at low concentrations [50]. A study involving gallate and PVP-coated AgNPs showed relatively high bacterial cell death rates

TABLE 3 Use of NPs in combination with current antibiotics and their mode of bacterial elimination as a possible basis for developing new formulations targeting the bacterial autophagic escape

Q2 Nanoparticle	Bacterial pathogen (MDR species)	Mode of action	Synergistic effect (antibiotics/ compounds)	Refs
Au	<i>Staphylococcus aureus</i>	Membrane disruption and inhibition of translation	5-Fluorouracil and gentamicin	[71–73]
Ag	<i>Pseudomonas aeruginosa</i> , <i>Klebsiella pneumoniae</i> , <i>Actinobacter baumannii</i>	Inhibition of translation, electron transport chain, cell wall synthesis and disruption of cell membrane, ROS generation	Penicillin G, amoxicillin, erythromycin, and vancomycin	[73–75]
Copper	<i>A. baumannii</i> , <i>P. aeruginosa</i> , <i>S. aureus</i>	ROS generation, lipid peroxidation, and DNA degradation	Tetracycline, rifampicin, chloramphenicol, gentamicin, and penicillin	[76]
Iron (oxide)	<i>Escherichia coli</i> , <i>K. pneumoniae</i>	ROS generation	Vancomycin	[74,77]
Zinc (oxide)	<i>Enterobacter aerogenes</i> , <i>E. coli</i> , <i>Klebsiella oxytolic</i> , <i>K. pneumoniae</i> , MRSA	ROS generation and membrane disintegration	Polymyxin B, ciprofloxacin, ceftazidime, and ampicillin	[78]
Silica	<i>S. aureus</i>	Membrane disruption	Bacitracin and kanamycin	[78,79]

compared with PVP or AgNPs alone [51]. These results were confirmed against carbapenem-resistant strains of *A. baumannii*. Another essential strategy is to prevent the enzymatic degradation of the antibiotics by integrating them with inhibitors; for example, incorporating clavulanic acid (an inhibitor of β -lactamase) with β -lactam antibiotics prevented their degradation by the resistant bacteria [52]. Knowledge of the administrative routes of the nanoformulation is essential because it eliminates the chances of clearance of the nanoformulations by serum proteins. Poly(lactic-co-glycolic acid) (PLGA) NPs used in conjugation with nitazoxanide results in a strong induction of autophagy during infections with *M. tuberculosis* [52]. Similarly, conjugation of acetylated dextran (ACE-DEX) materials also induce autophagy against *Salmonella* when used with an AR-12 [53]. Various autophagy modulators are either in preclinical trials or have been approved to treat various infections. For example, A-769662, an artificial compound that targets AMPK activation and, hence, modulates autophagy, is in preclinical trials [54]. Another antimicrobial treatment Q9 XXXXXX has also been approved to treat *Mycobacterium* infections because it alters the levels of reactive oxygen species (ROS) in the cells [55]. The molecular understanding of various bacterial infections could lead to the development of modulators of autophagy against specific bacteria. Table 4 provides details of the administrative routes of various drugs, such as liposomes, PLGA, and Ace-DEX, that exploit autophagic pathways.

Chimeric inhibitors of host–pathogen interactions

Phagocytes harboring ESKAPE pathogens, such as *S. aureus* and *K. pneumoniae*, act as ‘Trojan Horses’ to allow these intracellular bacteria to establish secondary infections [56]. *K. pneumoniae* engulfed by alveolar macrophages promote the activation of Akt to arrest phagosome maturation and avoid fusion to lysosomes. Evidence suggests that another ESKAPE pathogen, *S. aureus*, can invade and survive in phagocytes and has a mechanism to enable its escape from phagocytic vacuoles to the cytosol [56]. For intracellular pathogens, such as *Streptococcus* [57], identifying the surface adhesion molecules that interact with the human receptors is essential to recognize potential targets that could inhibit the initial host–bacterial interactions. For example, the presence of outer membrane protein OpcA in *Neisseria meningitidis* interacts with the human vitronectins, heparin sulfate, and proteoglycans. Fibronectin-binding protein (FnI) of *S. aureus* and *S. pyogenes* interacts with fibronectins, whereas internalin (InIB) of *L. mono-*

cytogenes interacts with human E cadherin, and the invasion protein of *Yersinia pseudotuberculosis* interacts with integrin [58]. Chimeric multiepitope vaccines could also be developed using comparative proteomics after identifying novel drug targets [59]. Another essential technique are gene silencing techniques, which can knock down the receptors necessary to internalize pathogens inside the host. Researchers have combined one such method to utilize nanoformulations encapsulating PDGFR- β siRNA (i.e., PDGFR- β siRNA-PEI-PLGA-PEG NPs) to induce autophagy and knockdown *PDGFRb* responsible for binding of *Chlamydia trachomatis* [1]. This nanoformulation induces autophagy and eliminates the gene responsible for the internalization of the bacterium. It might also be possible to target the known interaction between the outer membrane proteins Omp33-36 and OmpA of *A. baumannii* to effectively eliminate antibiotic-resistant strains of the pathogen [26], (Fig. 3). These interactions could be targeted because they impact the ability of pathogens to initiate autophagy and the establishment of persistent colonies inside the host cells [26]. Once we can inhibit the surface-level interactions of pathogens, it will be relatively straightforward to eliminate the risks of hijacking the genes responsible for the biogenesis of various lysosomal and autophagosomal genes [27]. For example, in *Acinetobacter*, internalization is initiated because the outer membrane protein induces the transcription of the lysosomal biogenesis gene, TF-EB, and, hence, the bacteria have a higher probability of establishing a reproductive niche inside the host [27]. Table 5 highlights the use of various hybrid molecules to enhance bacterial clearance from the host system.

Recombinant endolysins

Endolysins are a class of enzymes secreted by bacteriophages at the end of their replicative phase to degrade the bacterial cell membrane and spread infection [60]. Their ability to disrupt the bacterial cell membrane makes them efficient targets for antibacterial therapy [61]. Endolysins have two domains: a cell wall-binding domain and a enzymatically active domain [62]. Thus, they could be used alone or in combination with antibiotics to treat various bacterial infections. To increase their specificity, solubility, and bactericidal activity, there is scope to develop recombinant endolysins. For example, Ply187AN-KSH3b is a fusion of the two endolysins Ply187AN and LysK and showed tenfold higher antibacterial activity against *S. aureus* than either alone [61]. The synergy of such endolysins is conferred by the two disrupting

TABLE 4

Targeted bacterial pathogens and the mechanism of action of differential coatings used against pathogenic bacteria

Q3 Mode of delivery	Targeted microbes	Mechanism of action	Therapeutic route of administration	Refs
PLGA particles	<i>Mycobacterium tuberculosis</i> , <i>Escherichia coli</i>	Induction of autophagy in conjugation with nitazoxanide	Intranasal and intratracheal	[73–75]
Liposomes	<i>M. tuberculosis</i> , <i>Klebsiella pneumoniae</i> , <i>Pseudomonas aeruginosa</i> , <i>Staphylococcus aureus</i> , <i>Actinobacter baumannii</i>		Intranasal	[76]
Ace-DEX	<i>Salmonella typhimurium</i> , <i>Salmonella typhi</i>	Autophagy induction in combination with AR-12	Intranasal	[77]
Hydrogels	Methicillin-resistant <i>S. aureus</i>	Nitric oxide-releasing NPs	Skin wounds	[78]
SLN	<i>M. tuberculosis</i>	Sustained release of tobramycin and ciprofloxacin	Pulmonary based	[79]

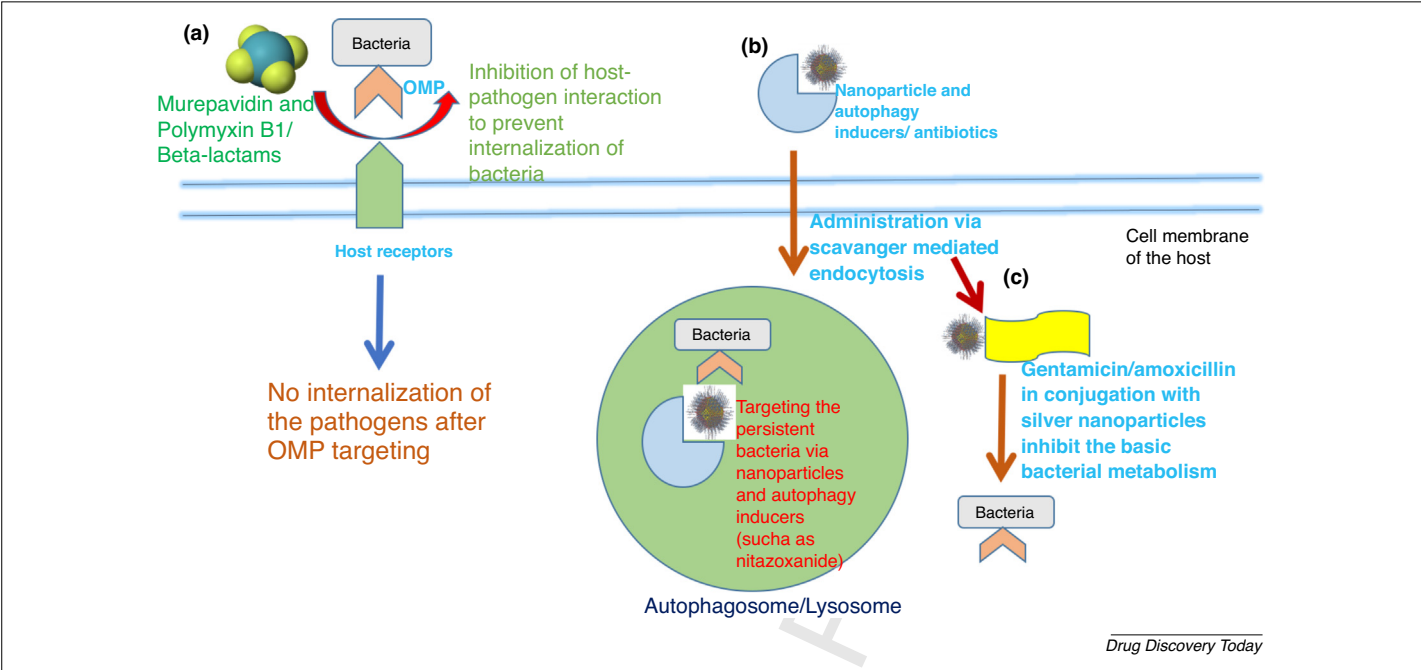


FIGURE 3 Administrative routes and targets of action suggested for therapeutics. (a) Inhibition of host–pathogen interactions by targeting outer membrane proteins with conjugated functional domains of antibiotics; (b,c) use of nanoparticles (NPs) in conjugation with autophagy inducers, such as nitazoxanide, and antibiotics, such as amoxicillin/gentamicin, respectively. These could be used specifically to target intracellular pathogens that show persistence inside the phagocytic vesicles or cytosol.

TABLE 5 Combinations of current antibiotics and drugs that might show efficacy against infections of ESKAPE pathogens when used with autophagy modulators

Q4 Combination of antibiotics	Compounds involved	Targets	Refs
β-Lactam–β-lactamase inhibitor	β-lactam amoxicillin and clavulanic acid (inhibitor of β-lactamase)	MDR Gram-positive and Gram-negative bacteria	[84]
Imipenem–cilastatin–relebactam triple combination	Carbapenem imipenem, cilastatin (inhibitor of carbapenemase), and relebactam (inhibitor of β-lactamase)	Gram-negative bacteria	[85,86]
Aspergillomarasmine A	Aspergillomarasmine A	Inhibits metallo-β-lactamases in <i>Klebsiella pneumoniae</i>	[87]
SPR741	SPR741–rifampin	Permeabilizes cell membrane of <i>Enterobacteriaceae</i> and <i>Actinobacter baumannii</i>	[88,89]
Antibiotic–antibiotic combination	Trimethoprim–sulfamethoxazole	<i>Enterobacteriaceae</i>	[90]

domains contained in a single 3D structure. Other examples include the combination of pneumococcal phage lysins Cpl-1 and Pal [60]. These endolysins could be used in synergy with inducers of autophagy to produce synthetic molecules that might result in enhanced bacterial clearance.

Challenges to developing new therapeutics

The main challenge to developing therapeutics against various pathogens lies in targeting different molecules responsible for autophagy escape. For example, Eis in *Mycobacterium* contains, VirA in *S. flexneri*, and SseF in *Salmonella* (see other examples in the Table 2) target different stages of autophagy in the host. A site and stage-specific approach has to be utilized to alter the levels of the bacteria surviving inside host cells. Effective clearance requires a

pathogen-specific approach. The use of molecules that degrade or silence these effector proteins used could help completely remove the bacteria. The evolution of offspring with beneficial traits compared with the parent generation has led to the evolution of genes that confer MDR to pathogenic bacteria, and can be achieved via genomic mutations, tolerance against the antibiotic treatments, and escaping the immune system of the host. Another main challenge lies with the administration of new therapeutics developed to eliminate infections. For example, NPs are administered intravenously and, as a result, whenever they come in contact with the serum, the plasma proteins coat them and prepare them for opsonization. The mononuclear phagocyte system (MPS) prevents them from reaching their targets, as demonstrated by Feby *et al.* [63]. By contrast, it will be important to

evaluate the chimeric/hybrid molecules for their long-term effects on the physiological process of the host machinery. The monitoring of new therapeutic drugs after administration is necessary before incorporating them into current treatments to minimize the risk of developing new types of resistant strains and to confer a maximum benefit to the host.

Concluding remarks and future perspectives

Given the decline in the clearance of bacterial infections with currently available treatments, it has become essential to develop new treatment strategies. Here, we have discussed the various escape strategies used by different pathogens to outperform the host immune system. The onset of a bacterial infection initiates autophagy in the host. However, even when the pathogen has been internalized, they are able to secrete various effectors that help to establish a reproductive niche inside host cells, resulting in the onset of persistent bacterial infections. Therefore, we have focused on reviewing new therapeutics, such as autophagy indu-

cers in synergism with various existing antibiotics. Such nanoformulations have shown antibacterial effects against various MDR bacterial pathogens. PLGA and AceDEX-coated NPs/antibiotics are particularly promising because they exploit the autophagic route. The challenges that need addressing with such new therapeutics are the mechanisms of action after their administration and their clearance by the host system. Hence, the synergy of the combinations of therapeutics and the activation of the immune system of the host could help combating MDR species without the risk of developing new types of resistance.

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